

12(a)	(minimum) energy required to separate the nucleons	M1
	to infinity	A1
12(b)(i)	37 2	B1
12(b)(ii)	fission	B1
12(b)(iii)	binding energy per nucleon smaller for U than for Cs	B1
12(c)	Current ratio 2 Y to 1 Zr, so initially 3 Y $2 = 3 e^{-\lambda t}$ $\lambda = 0.693 / 2.7$	C1
	$\ln(2 / 3) = - (\ln 2 / 2.7)t$	C1
	$t = 1.6$ days	A1
	<i>or</i>	
	$(\frac{1}{2})^n = 2 / 3$	(C1)
	$n = 0.585$	(C1)
	time = 0.585×2.7 = 1.6 days	(A1)